

Master's Theorem

MODULE I

Master's Theorem

- It is used to analyse recurrence relation.
- $T(n) = aT\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$,
 $a \geq 1, b > 1, k \geq 0$ and p is real number
- Case 1: if $a > b^k$, then $T(n) = \theta(n^{\log_b a})$
- Case 2: if $a = b^k$, then
 - a) if $p > -1$, then $T(n) = \theta(n^{\log_b a} \log^{p+1} n)$
 - b) if $p = -1$, then $T(n) = \theta(n^{\log_b a} \log^2 n)$
 - c) if $p < -1$, then $T(n) = \theta(n^{\log_b a})$
- Case 3: if $a < b^k$, then
 - a) if $p \geq 0$, then $T(n) = \theta(n^k \log^p n)$
 - b) if $p < 0$, then $T(n) = \theta(n^k)$

Example Problems:

1. $T(n) = 3T\left(\frac{n}{2}\right) + (n^2)$

2. $T(n) = 4T\left(\frac{n}{2}\right) + (n^2)$

3. $T(n) = T\left(\frac{n}{2}\right) + (n^2)$

4. $T(n) = 2^n T\left(\frac{n}{2}\right) + (n^n)$

5. $T(n) = 2T\left(\frac{n}{2}\right) + (n \log n)$

6. $T(n) = 2T\left(\frac{n}{2}\right) + n/\log n$

7. $T(n) = 2T\left(\frac{n}{4}\right) + (n^{0.51})$

8. $T(n) = 16T\left(\frac{n}{4}\right) + (n)$

9. $T(n) = 0.5T\left(\frac{n}{2}\right) + (1/n)$

10. $T(n) = 6T\left(\frac{n}{3}\right) + (n^2 \log n)$

11. $T(n) = 64T\left(\frac{n}{8}\right) - (n^2 \log n)$

12. $T(n) = 7T\left(\frac{n}{3}\right) + (n^2)$

13. $T(n) = 4T\left(\frac{n}{2}\right) + cn$

14. $T(n) = 4T\left(\frac{n}{2}\right) + (\log n)$

15. $T(n) = 3T\left(\frac{n}{2}\right) + n$

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1. $T(n) = 3T\left(\frac{n}{2}\right) + (n^2)$
2. $T(n) = 4T\left(\frac{n}{2}\right) + (n^2)$
3. $T(n) = T\left(\frac{n}{2}\right) + (n^2) \rightarrow \theta(n^2)$
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Example Problems:

1. $T(n) = 3T\left(\frac{n}{2}\right) + (n^2)$

a = ?, b = ?, k = ?, p = ?

Example Problems:

$$1. T(n) = 3T\left(\frac{n}{2}\right) + (n^2)$$

$$a = 3, b = 2, k = 2, p = 0$$

(conditions $a \geq 1, b > 1, k \geq 0$,
 p is a real number)

$$\begin{array}{cc} a & b^k \\ 3 & 2^2 \end{array}$$

$$3 < 4 \Rightarrow a < b^k$$

Falls under case 3a

$$\begin{aligned} p \geq 0 &\rightarrow T(n) = \theta(n^k \log^p n) \\ &= \theta(n^2 \log^0 n) \\ &= \theta(n^2) \end{aligned}$$

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Example Problems:

1. $T(n) = 4T\left(\frac{n}{2}\right) + (n^2)$

$a = 4, b = 2, k = 2, p = 0$

(conditions $a \geq 1, b > 1, k \geq 0$,
 p is a real number)

a	b^k
4	2^2

$$4 = 4 \Rightarrow a = b^k$$

Falls under case 1a

$$\begin{aligned} p > -1 &\Rightarrow T(n) = \theta(n^{\log_b a} \log^{p+1} n) \\ &= \theta(n^{\log_2 4} \log^{0+1} n) \\ &= \theta(n^2 \log n) \end{aligned}$$

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